STA 332 Blower (S.O.O. 6379) – General Data

1. General

Provisions have been made for the installation of a single electrically-driven centrifugal (squirrel-cage) style blower, on the STA 332 bulkhead. The blower directs air forward into the passenger compartment, to aid in air circulation.

The blower is mounted to a formed sheet metal panel that is fastened by screws to the aft face of the STA 332 bulkhead. On aircraft with a closed STA 332 bullhead, air is moved through a sheet metal duct and exits through a screen. On aircraft with an open STA 332 bulkhead, the screen is fastened directly to the filler panel and no duct is required. The blower has two speeds: high speed provides 28 volt DC directly to the blower motor, and low speed is provided through a 3 ohm resistor located on the blower housing.

The blower fan is operated by a CABIN BLOWER STA 332 switch located in the flight compartment, above and behind the Pilot on the STA 111 bulkhead.

28 Volt DC power is provided from the right bus through a 15 amp CABIN FANS (B14) circuit breaker on the main circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

2. Adjustment/Test

A. Operational Test

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Select CABIN BLOWER STA 332 switch to HIGH and LOW, observing the operation at each setting.

Note The fan motor should be smooth running without irregular noise, rattling, or vibration.

3) Select CABIN BLOWER STA 332 switch to OFF.

4) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

3. Removal/Installation

A. Remove STA 332 Blower

1) Ensure aircraft power is OFF. (Refer to 24-00-00, External Power – General Data).

2) Pull the CABIN FANS (B14) circuit breaker from main circuit breaker panel and fit safety clip.

3) Remove the outlet screen mounting screws from the forward side of STA 332 bulkhead.

4) If installed, remove the duct assembly by removing the four (4) screws attaching the duct assembly to the blower assembly and filler panel assembly.

5) Remove two (2) screws securing blower assembly to filler assembly (upper holes).

6) Carefully pull the blower from the mounting assembly, and disconnect the blower electrical wiring.

7) Remove the blower assembly from the aircraft.

B. Install STA 332 Blower

1) Ensure aircraft power is OFF. (Refer to 24-00-00, External Power – General Data).

2) Ensure CABIN FANS (B14) circuit breaker is pulled and fitted with safety clip.

3) Examine the blower assembly for damage, replace if necessary. Ensure the outlet screen is clean of debris.

4) Place the blower assembly near the mounting bracket and connect the electrical wiring.

5) Install two (2) screws in upper holes of filler panel to secure blower assembly.

6) If equipped, install the duct assembly by installing the four (4) screws attaching the duct assembly to the blower assembly and filler panel assembly.

7) Install the mounting screws for the outlet screen at the forward side of STA 332 bulkhead.

8) Ensure the blower assembly is secure on mounting bracket.

9) Remove safety clip and close CABIN FANS (B14) circuit breaker on the main circuit breaker panel.

10) Do an operational test as per Para 2.A.

4. Servicing

A. Blower Failure

Note The blower fan is not repairable. If a blower fan is unserviceable, it must be replaced.

1) Ensure aircraft power is OFF. (Refer to 24-00-00, External Power – General Data).

2) Disconnect the wiring splices at the blower.

3) Remove the outlet screen.

4) If the aircraft has a closed STA 332 bulkhead, remove the duct.

5) Remove the screws fastening the blower fan assembly to the filler panel.

6) Replace the blower fan assembly with a new part.

7) Connect the new blower fan to the wiring splices and re-install screws, duct (if equipped) and outlet screen (refer to Para 3.B).

5. Troubleshooting

A. Fan Fault

1) Ensure CABIN FANS (B14) circuit breaker is closed.

2) Ensure the CABIN BLOWER STA 332 switch is in the LOW or HIGH position.

3) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.